

Graph Coloring with Neutral Atoms: Assignment Report

Hubert Hugo, Favre Lucas, Le Roux Florian

Contents

1	Introduction	1
2	Step 1: Problem Modeling	1
3	Step 2: Algorithm Design	2
3.1	Maximum Independent Set problem	3
3.2	Overview of the Adiabatic Evolution Method	3
3.3	Implementation Details	3
4	Step 5: Report and Justification	5
4.1	Problem Modeling	5
4.2	Algorithm Choice	5
4.3	Future Improvements	5
5	Conclusion	5

1 Introduction

A telecommunication company has tasked us with addressing the problem of assigning radio frequencies to a set of antennas distributed over a geographical area. Interference occurs when two antennas within a certain proximity use the same frequency. The objective is to minimize the number of frequencies while ensuring minimal interference.

2 Step 1: Problem Modeling

The frequency assignment problem can be modeled as a **graph coloring problem on unit-disk graphs**. Here's how:

- Each antenna is represented as a **vertex** in the graph.
- An **edge** is drawn between two vertices if the corresponding antennas are within the interference radius (8.7 km).
- A **valid coloring** of the graph assigns a frequency (color) to each vertex such that no two connected vertices share the same color.

- The goal is to minimize the number of colors used (frequencies assigned).

The physical positions of the antennas define the graph as follows:

- | | |
|-------------------------|-------------------------|
| • Antenne 1 : (0, 0) | • Antenne 5 : (9, 0) |
| • Antenne 2 : (3, 5.2) | • Antenne 6 : (9, 5.2) |
| • Antenne 3 : (6, 0) | • Antenne 7 : (9, 10.4) |
| • Antenne 4 : (9, -5.2) | • Antenne 8 : (12, 0) |

Using these positions, edges are determined based on the Euclidean distance between antennas. For example:

- Antenna 1 and Antenna 2 are connected because their distance is approximately 6 km, which is within the interference radius.
- Antenna 1 and Antenna 3 are also connected because their distance is 6 km.
- Antenna 3 and Antenna 8 are connected because their distance is also 6 km.
- Antenna 4 and Antenna 5 are not connected because their distance is exactly 5.2 km, less than the interference radius.

Example with Graph Coloring

Below is an example of a possible coloring for this graph:

- Assign **Red** to Antenna 1.
- Assign **Blue** to Antenna 2 and Antenna 3.
- Assign **Green** to Antenna 4.
- Assign **Yellow** to Antenna 5 and Antenna 6.
- Assign **Purple** to Antenna 7.
- Assign **Cyan** to Antenna 8.

This coloring satisfies the interference constraints, as no two connected vertices share the same color. A visual representation of the graph and coloring can help illustrate the relationships and assignments. However, the solution proposed here is far from optimal, and the goal of our code will be to address this issue.

3 Step 2: Algorithm Design

We propose an **adiabatic evolution method** to solve the graph coloring problem. This quantum algorithm exploits the principles of adiabatic quantum computing. To help us with this method, we will use the MIS algorithm as shown in the GitHub course.

3.1 Maximum Independent Set problem

The Maximum Independent Set (MIS) problem is a combinatorial optimization problem in graph theory. Given an undirected graph with vertices and edges, the goal of the MIS problem is to find the largest subset of vertices such that no two vertices in the subset are adjacent.

The MIS problem has several applications, including network design and resource allocation. This problem is also known to be NP-hard, as exact algorithms are impractical for large graphs. By using quantum computers, it may be possible in the future to solve this problem through adiabatic quantum computing methods.

3.2 Overview of the Adiabatic Evolution Method

Adiabatic quantum computing relies on the **adiabatic theorem**, which states that a quantum system remains in its ground state if changes to its Hamiltonian are sufficiently slow. The steps are as follows:

1. Define an **initial Hamiltonian** H_0 with a known, simple ground state.
2. Define the **problem Hamiltonian** H_p , whose ground state encodes the solution to the graph coloring problem.
3. Evolve the system from H_0 to H_p over time T using the interpolation:

$$H(t) = (1 - s(t))H_0 + s(t)H_p,$$

where $s(t)$ is a time-dependent schedule varying smoothly from 0 to 1.

3.3 Implementation Details

To implement the adiabatic algorithm for graph coloring with neutral atoms, the following steps are followed:

Neutral Atom Representation

Each vertex in the graph is represented by a neutral atom trapped in an optical tweezer or lattice. The spatial arrangement of the atoms corresponds to the graph structure. Interactions between atoms encode the graph edges, ensuring that connected vertices influence each other during the computation.

Hamiltonian Definitions

- Before starting the computation, we'll need an initial Hamiltonian H_0 . This will represent all the atoms in a non-colored state (inactive).

$$H_0 = \sum_i \Delta_i n_i + \sum_{i,j} V_{ij} n_i n_j,$$

With Δ_i a detuning parameter, n_i the occupation number of the i -th atom, and V_{ij} the interaction between atoms i and j .

- Next we'll need to encode the graph coloring constraints. That we can do by using another Hamiltonian H_p , that penalizes configurations where adjacent vertices, here atoms, share the same color.

$$H_p = \sum_i \delta_i n_i - \sum_{i,j} W_{ij} n_i n_j,$$

With δ_i encouraging activation (coloring), and W_{ij} penalizing shared colors between connected vertices.

Adiabatic Evolution

Step 1: Initialization

The system begins in the ground state of H_0 , with all atoms uncolored.

Step 2: Evolution

Over a time T , the Hamiltonian evolves slowly from H_0 to H_p using the schedule:

$$H(t) = (1 - s(t))H_0 + s(t)H_p,$$

where $s(t)$ is a smooth function of time.

Step 3: Final State

At the end of the evolution, the system is in the ground state of H_p , representing a valid graph coloring.

Iterative Process for Coloring

The algorithm proceeds iteratively:

- After finding a maximum independent set (MIS) corresponding to a color, the vertices in the MIS are removed from the graph.
- Then, the process will find the MIS of the reduced graph.
- Until having a two vertices graph, the process must repeat.
- At a two vertices graph, each vertices is attributed a color

This iterative approach ensures that the minimum number of colors is used while adhering to the graph's constraints.

4 Step 5: Report and Justification

4.1 Problem Modeling

The use of unit-disk graphs effectively captures the spatial interference between antennas. Graph coloring offers a structured mathematical framework for frequency assignment.

4.2 Algorithm Choice

The adiabatic evolution method is chosen for its ability to handle combinatorial optimization problems like graph coloring. Its compatibility with neutral atom platforms ensures feasibility in real-world quantum computing setups.

4.3 Future Improvements

Potential enhancements include incorporating error correction mechanisms and optimizing the adiabatic schedule to improve computational efficiency.

5 Conclusion

This report outlines the modeling and algorithmic approach to solving the antenna frequency assignment problem using graph coloring and adiabatic quantum computing. The proposed method leverages quantum principles and neutral atom technologies to provide an innovative solution.